5.13 Solution: The current density is given by

$$\begin{aligned}\mathbf{J}(r,\theta,\phi) &= \sigma\omega r\sin\theta\,\delta(r-a)\,\hat{\boldsymbol{\phi}}\\ &= \sigma\omega r\sin\theta\,\delta(r-a)\left(-\sin\phi\,\hat{\mathbf{i}}+\cos\phi\,\hat{\mathbf{j}}\right).\end{aligned}$$

The vector potential can be calculated as

$$\begin{aligned}\mathbf{A}(\mathbf{r}) &= \frac{\mu_0}{4\pi}\int\frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d^3r'\\ &= \frac{\mu_0}{4\pi}\int_0^\infty r'^2dr'\int_{-1}^{1}d(\cos\theta')\int_0^{2\pi}d\phi'\;\sigma\omega r'\sin\theta'\,\delta(r'-a)\left(-\sin\phi'\,\hat{\mathbf{i}}+\cos\phi'\,\hat{\mathbf{j}}\right)\\ &\quad\times\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\frac{4\pi}{2l+1}Y_{lm}(\cos\theta)\,Y^*_{lm}(\cos\theta')\,e^{im(\phi-\phi')}\,\frac{r_<^l}{r_>^{l+1}}.\end{aligned}$$

Only $l=1$, $m=\pm1$ terms contribute,

$$\begin{aligned}\mathbf{A}(\mathbf{r}) &= \frac{\mu_0}{4\pi}\sigma\omega a^3\int_{-1}^{1}d(\cos\theta')\int_0^{2\pi}d\phi'\;\sin\theta'\left(-\sin\phi'\,\hat{\mathbf{i}}+\cos\phi'\,\hat{\mathbf{j}}\right)\\ &\quad\times\frac{1}{2}P_1^1(\cos\theta)\,P_1^1(\cos\theta')\,2\cos(\phi-\phi')\,\frac{r_<}{r_>^2}\\ &= \frac{\mu_0}{4}\sigma\omega a^3\sin\theta\left(-\sin\phi\,\hat{\mathbf{i}}+\cos\phi\,\hat{\mathbf{j}}\right)\int_{-1}^{1}\sin^2\theta' d(\cos\theta')\,\frac{r_<}{r_>^2}\\ &= \frac{\mu_0\sigma\omega a^3}{3}\frac{r_<}{r_>^2}\,\sin\theta\,\hat{\boldsymbol{\phi}}.\end{aligned}$$

Therefore, for $r>a$,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0\sigma\omega a^4}{3r^2}\sin\theta\,\hat{\boldsymbol{\phi}} = \frac{\mu_0\sigma a^4}{3r^3}\,\boldsymbol{\omega}\times\mathbf{r},$$

and, for $r<a$,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0\sigma\omega a}{3}\,r\sin\theta\,\hat{\boldsymbol{\phi}} = \frac{\mu_0\sigma a}{3}\,\boldsymbol{\omega}\times\mathbf{r}.$$

To calculate the magnetic induction, by definition

$$\mathbf{B}(\mathbf{r}) = \nabla\times\mathbf{A}(\mathbf{r}),$$

we can show that, for $r<a$,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0\sigma a}{3}\nabla\times(\boldsymbol{\omega}\times\mathbf{r}) = \frac{\mu_0\sigma a}{3}\left[\boldsymbol{\omega}(\nabla\cdot\mathbf{r})-(\boldsymbol{\omega}\cdot\nabla)\mathbf{r}\right] = \frac{2\mu_0\sigma a}{3}\,\boldsymbol{\omega},$$

and for $r>a$,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0\sigma a^4}{3}\nabla\times\left(\boldsymbol{\omega}\times\frac{\mathbf{r}}{r^3}\right) = \frac{\mu_0\sigma a^4}{3}\left[\boldsymbol{\omega}\left(\nabla\cdot\frac{\mathbf{r}}{r^3}\right)-(\boldsymbol{\omega}\cdot\nabla)\frac{\mathbf{r}}{r^3}\right] = \frac{\mu_0\sigma a^4}{3}\left[\frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\boldsymbol{\omega})-\boldsymbol{\omega}}{r^3}\right].$$